

Inflation, Unemployment, and Lending Relationships

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Abstract

We show that inflation and unemployment can exhibit a hump-shaped relationship, focusing on a novel financial intermediation channel: higher inflation raises the return to bank–firm lending relationships, drawing firms into credit markets and boosting their capital investment. This channel competes with the liquidity channel of Berentsen et al. (2011), in which the inflation tax compresses goods-market trade and depresses vacancy creation. Below an endogenous threshold the liquidity channel dominates and unemployment rises with inflation; above it the credit channel activates and eventually prevails, pulling unemployment down. Unlike the monotone prediction of Berentsen et al. (2011), the interplay between the two channels generates a non-monotone inflation–unemployment locus. The result is robust to endogenous relationship fragility and cross-country variation in financial development.

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1. Introduction

We revisit the long-run relationship between inflation and unemployment through a novel channel: the endogenous formation of bank–firm lending relationships. In the canonical framework of Berentsen et al. (2011), inflation raises unemployment monotonically because the inflation tax compresses goods-market trade and reduces vacancy creation. We show that once firm-side credit markets are introduced, a countervailing force emerges that can overturn this monotone result and generate a hump-shaped inflation–unemployment locus.

To formalize this idea, we build a general equilibrium model with three integrated markets. Households and firms trade consumption goods for money in a decentralized goods market à la Lagos and Wright (2005), where bilateral matching and take-it-or-leave-it offers determine the real value of money balances. Firms and workers match in a frictional labor market with free entry of vacancies, following Berentsen et al. (2011) (henceforth BMW). On top of this, firms and banks form lending relationships through directed search: banks post credit contracts, firms apply, and matched pairs finance capital investment at terms determined by Nash bargaining, as in Bethune et al. (2021). The three markets interact through firms’ match values, which depend on goods-market revenues, credit access, and vacancy-creation incentives.

This model structure gives rise to two competing forces. The key addition to the BMW framework is a *financial intermediation channel*: higher inflation raises the nominal spread, making lending relationships more valuable to firms, which attracts bank entry and expands the fraction of firms with access to credit. This channel competes directly with the *liquidity channel* of BMW, in which the inflation tax compresses household money balances, reduces goods-market trade, and depresses vacancy creation.

The financial intermediation channel operates through two margins. On the extensive margin, a higher spread raises the surplus from a bank–firm match, attracting bank entry and expanding the banking fraction. On the intensive margin, the optimal credit contract gives banked firms a capital advantage that is strictly increasing in the spread: above an endogenous threshold, banked firms’ capital diverges sharply from that of unbanked firms.

Below this threshold, the credit line is too small to expand firms’ capital beyond the self-financed optimum—banked and unbanked firms hold identical capital, and the financial intermediation channel contributes only a second-order cash-saving effect. The liquidity channel of Berentsen et al. (2011) therefore dominates, and unemployment rises with inflation as in the standard model. Above the threshold, the capital-investment margin activates: both the banking fraction and the per-firm capital gap grow with inflation, amplifying the financial intermediation channel until it overtakes the liquidity drag. The unemployment peak marks this transition.

We calibrate the model to standard U.S. targets and confirm that the hump-shaped locus emerges quantitatively. Below the threshold, the liquidity channel dominates: the inflation tax erodes household money balances, compressing goods-market trade and reducing the surplus that sustains job creation. Unemployment rises. Above the threshold, the credit channel prevails: the rising nominal spread draws banks into the market, expands the fraction of firms with credit access, and raises aggregate capital investment sufficiently to pull unemployment back down. The calibrated locus is non-monotone, consistent with the theoretical prediction.

Two extensions explore the robustness of the hump-shaped locus. First, endogenous relationship fragility: if inflation erodes lending relationships, the banking fraction peaks and then declines, bounding the credit channel. At moderate fragility the hump shape is preserved; at extreme fragility the locus becomes N-shaped, with unemployment eventually rising again as relationships dissolve. Second, financial development: lower monitoring costs reduce the regime threshold, activating the credit channel at lower inflation while leaving the Friedman-rule outcome unchanged. The two extensions are complementary: monitoring costs determine *when* the credit channel activates, while fragility determines *how far* it can operate.

This paper makes three contributions. First, we provide the first model in which the long-run inflation–unemployment relationship is non-monotone due to the endogenous formation of bank–firm lending relationships. Unlike existing mechanisms that rely on smooth reversals of a single force, the non-monotonicity here arises from the discrete activation of a new channel—financial intermediation—at an endogenous inflation threshold. Second, we characterize this threshold analytically and show that it partitions the policy space into a liquidity-dominated regime and a credit-dominated regime, each with distinct comparative statics. Third, the extensions to fragility and financial development demonstrate that the hump shape is robust and identify the separate roles of relationship durability and entry costs in shaping the locus.

Related literature. This paper builds on the New Monetarist program surveyed by Lagos et al. (2017). Lagos and Wright (2005) introduce the alternating market structure—a decentralized goods market with bilateral search followed by a frictionless centralized market with quasi-linear preferences—that provides the standard microfoundation for monetary theory. Rocheteau and Wright (2005) compare Nash bargaining, competitive pricing, and directed search in this framework, showing that the welfare effects of inflation depend critically on market structure. Extensions explore the interaction of money with capital (Aruoba et al. 2011) and with credit (Gu et al. 2016); the latter show that money and credit are not co-essential in frictionless environments. Our model departs from this irrelevance result because bank credit is not a frictionless substitute—firms must endogenously form lending relationships at a cost, and only a fraction of firms gain access

to credit in equilibrium. Berentsen et al. (2011) embed the Lagos–Wright goods market into a Mortensen–Pissarides labor market and establish that the inflation tax on decentralized trade generates a positive long-run relationship between inflation and unemployment. We extend BMW by introducing a firm-side credit market that produces a countervailing employment channel.

The canonical positive long-run relationship between inflation and unemployment established by Berentsen et al. (2011) has been revisited by several recent contributions. Dong (2011) shows that the sign can reverse under competitive search (directed price posting) when sellers have sufficient market power. Gomis-Porqueras et al. (2020) extend BMW to include capital accumulation and find a hump-shaped relationship between inflation and capital, driven by competing extensive-margin (entry) and intensive-margin (investment) effects. Ait Lahcen et al. (2022) document state-dependent and nonlinear unemployment effects of the inflation tax in a calibrated model with aggregate shocks, though their nonlinearity is in the magnitude rather than the sign of the effect. The papers most closely related to ours are Bethune et al. (2024) and Gu et al. (2025). Bethune et al. generate a non-monotone Phillips curve through workers’ liquidity-based outside options: at low inflation, wage moderation tightens the labor market, but at high inflation firms’ financing costs dominate. Gu, Jiang, and Wang construct a model with imperfect credit and derive two opposing channels—a wage-bargaining channel and a cash-financing channel—whose relative strength depends on credit conditions. Our mechanism differs from both: the non-monotonicity arises from the *endogenous activation* of a discrete financial intermediation channel at a threshold inflation rate, driven by bank entry and lending relationship formation, rather than from a smooth reversal of a single force.

The financial side of our model draws on the lending-relationship literature. Petersen and Rajan (1994) provide the foundational empirical evidence that lending relationships generate quantitatively important financing advantages for small businesses: firms with closer bank ties face lower borrowing costs and obtain credit more readily. Rocheteau et al. (2018) introduce a New Monetarist framework for corporate finance in which entrepreneurs finance investment via trade credit, bank assets, or currency in OTC loan markets, and derive pass-through conditions for monetary policy to real lending rates. Bethune et al. (2021) build on Rocheteau et al. by adding endogenous bank–firm matching with free entry, deriving the key result that the value of a banking relationship is increasing in the nominal spread. Their analysis focuses on optimal monetary policy following a banking crisis and abstracts from labor markets. We introduce Mortensen–Pissarides frictions into the BRZW framework and show that the inflation-driven expansion of banking has first-order effects on steady-state unemployment through average firm productivity. Bethune et al. (2015) extend Mortensen–Pissarides with household credit frictions and show that borrowing constraints reduce aggregate vacancy posting; our financial chan-

nel differs in operating on the firm side through investment financing rather than on household balance sheets.

The paper also relates to the literature combining financial and labor market frictions. Wasmer and Weil (2004) is the seminal contribution embedding search frictions in both credit and labor markets: entrepreneurs must search for external finance before posting vacancies, so tighter credit raises the effective cost of job creation and amplifies unemployment. Petrosky-Nadeau and Wasmer (2013) extend this framework to a stochastic DMP setting and show that financial frictions amplify the volatility of labor market tightness, providing a partial resolution of the Shimer puzzle. Our model shares the spirit of Wasmer–Weil—financial access matters for job creation—but differs fundamentally in the friction modeled. In Wasmer–Weil, the friction is credit *availability* (whether a firm obtains a loan); in our model, it is credit *depth* (banked firms invest more capital), so the channel runs through average firm productivity rather than the fixed cost of vacancy creation.

Road map. Section 2 presents the model. Section 3 characterizes the equilibrium, establishes analytical results on the two channels, and presents quantitative results. Section 4 extends the model to endogenous relationship fragility and cross-country variation in financial development. Section 5 concludes.

2. The Model

This section describes the environment and the decision problems of each agent. Section 2.1 lays out the timing, agents, and market structure. Section 2.2 presents the household problem, including DM trade and the labor market transition. Section 2.3 characterizes firms’ capital investment, goods-market participation, and vacancy posting. Section 2.4 describes the bank–firm credit market with endogenous relationship formation. Section 2.5 collects the equilibrium conditions and establishes existence and uniqueness.

2.1. Environment

Time and agents. Time is discrete and infinite. The economy is populated by a unit measure of households, a continuum of firms (free entry), a continuum of banks (free entry), and a competitive fringe of capital-good suppliers.

Households are either *employed* (measure $1 - u$) or *unemployed* (measure u). Firms are either *banked* (measure ℓ , maintaining a lending relationship with a bank) or *unbanked* (measure $1 - \ell$). Both u and ℓ are endogenous and determined in steady state.

Goods and money. There are three goods: a capital good k (storable within a period but not across periods), a consumption good q traded in decentralized markets, and a numeraire good x traded in the centralized market. A storable, intrinsically worthless fiat

currency grows at gross rate $\mu \geq \beta$, where $\beta \in (0, 1)$ is the quarterly discount factor. In stationary equilibrium $1 + i = \mu/\beta$, where i is the annualized net nominal interest rate and s is the nominal spread.

All agents have quasi-linear preferences over the numeraire good (Lagos and Wright 2005), which delivers a degenerate distribution of money holdings at the start of each period: every agent exits the centralized market with identical real balances *within* each agent type (households, unbanked firms, banked firms each carry distinct optimal portfolios).

Anonymity and means of payment. In the two decentralized markets—the capital-good market (DM₁) and the consumption-good market (DM₂)—trading partners meet anonymously and cannot enforce future promises against each other. Consequently, firm IOUs are not accepted as means of payment: a firm’s creditworthiness is private information unverifiable by trading counterparties. Only fiat money and bank-certified credit serve as payment, as in Lagos and Wright (2005) and Bethune et al. (2021). This makes the banking fraction ℓ the sole margin through which credit enters decentralized exchange.

Period structure. Each period consists of five sequential stages:

Stage	Market	Participants
1	Labor market (MP)	Households + firms
2	Bank–firm search	Unbanked firms + banks
3	DM ₁ : capital market	Firms (buyers) + capital suppliers
4	DM ₂ : goods market	Households (buyers) + employed firms
5	AD: centralized market	All agents

Bank–firm matching (Stage 2) precedes capital investment (Stage 3), so a firm’s banking status is determined *before* it chooses its capital holdings. This ordering reflects the empirical finding that lending relationships are established prior to investment decisions (Petersen and Rajan 1994).

Markets. Each stage hosts a distinct market with its own frictions and means of exchange.

Stage 1 — Labor market (MP). Firms post vacancies at flow cost c_v . Households and firms match through a constant-returns matching function $\mathcal{M}(u, v) = Zu^{1-\sigma_m}v^{\sigma_m}$, where u is unemployment and v is the vacancy rate. Upon a match, the firm and worker bargain over the wage; unmatched matches separate exogenously at rate δ_L per period.

Stage 2 — Bank–firm credit market. Unattached banks pay an entry cost κ and search for unbanked firms; matches form at rate $\alpha_b(\theta_b)$, where θ_b is credit-market tightness (bank vacancies relative to unbanked firms). Existing relationships dissolve at the exogenous rate δ_b per period. Banks have monitoring capacity that allows them to certify and enforce firm credit, which ordinary trading partners cannot do.

Stage 3 — DM₁: capital-good market. Firms purchase capital goods from competitive suppliers (linear technology $k = h$, so the price of capital equals one). An unbanked firm finances its capital entirely with its own cash holdings m_u . A banked firm supplements cash with bank credit d_b arranged in Stage 2, enabling it to acquire up to $k_b = m_b + d_b$ units of capital.

Stage 4 — DM₂: consumption-good market. All households spend their cash holdings to purchase consumption goods q from employed firms. Buyers and sellers are matched bilaterally; trading partners are anonymous, so only cash is accepted as payment. Terms of trade (q, d) are determined by generalized Nash bargaining.

Stage 5 — AD: centralized market. All agents trade the numeraire good x , settle financial obligations, and choose money balances z' to carry into the next period. Quasi-linear preferences in x imply that every agent exits with identical real balances, delivering the degenerate money distribution that underlies the Lagos–Wright framework.

Monetary policy. The government issues fiat money and expands the money supply at gross rate $\mu \geq \beta$ per period; new money is rebated lump-sum to households. In steady state, the **Fisher equation** relates the nominal rate i , the growth rate μ , and the discount factor β :

$$(1) \quad 1 + i = \frac{\mu}{\beta}.$$

The government's sole policy instrument is $i \geq 0$; the Friedman rule corresponds to $i = 0$ ($\mu = \beta$).

2.2. Households

A household's employment status is $e \in \{0, 1\}$. Employed households ($e = 1$) earn wage w ; unemployed households ($e = 0$) receive benefit b . Preferences are quasi-linear: period utility is $x + v(q)$, where x is numeraire consumption and $v(q)$ is utility from consuming q in DM₂ ($v' > 0$, $v'' < 0$, $v'(0) = \infty$).

In the AD market, a household with real balances z and status e solves

$$W_h^e(z) = \max_{z' \geq 0} \{I^e + z - z' + \beta U_h^e(z')\},^1$$

where $I^e = ew + (1 - e)b + \Pi - T$, with Π dividends and T a lump-sum tax. Since W_h^e is linear in z , the optimal z' is independent of (z, e, I^e) : every household exits the AD market with identical real balances (Lagos and Wright 2005).

¹ z' denotes real balances at the current price level chosen in the AD market. The inflation cost enters through the next DM subperiod: the z' units are worth $z'/(1+\pi)$ in real goods at the start of next period. Differentiating with respect to z' gives the first-order condition $\beta \partial U_h^e(z^*)/(1+\pi) = 1$, or equivalently $(1+i) \beta \partial U_h^e(z^*) = 1$ via the Fisher equation.

In DM₂, all households (employed and unemployed) match bilaterally with employed firms. With 1 buyer (the unit mass of households) and $1 - u$ sellers (employed firms), the meeting rates are

$$\begin{aligned}\alpha_h(u) &\equiv Z(1 - u)^{\sigma_m}, \\ \alpha_f(u) &\equiv Z(1 - u)^{\sigma_m - 1}.\end{aligned}$$

Following Liu (2008), we evaluate these meeting probabilities at the steady-state unemployment rate $\bar{u}$, replacing $\alpha_h(u)$ and $\alpha_f(u)$ by constants:

$$\bar{\alpha}_h \equiv Z(1 - \bar{u})^{\sigma_m}, \quad \bar{\alpha}_f \equiv Z(1 - \bar{u})^{\sigma_m - 1}.$$

With α_h held fixed, the LW condition determines $q^*(i)$ independently of u .

A household entering DM₂ with real balances z has value

$$V_h^e(z) = \alpha_h(u)[v(q) - d] + W_h^e(0) + z,$$

using the linearity of W_h^e . Upon a bilateral match, the household and firm bargain via generalized Nash bargaining:

$$(NB-h) \quad \max_{q, d} [v(q) - d]^{\eta_h} [d - q]^{1 - \eta_h} \quad \text{s.t.} \quad d \leq z,$$

where $c(q) = q$ (linear production cost) and $\eta_h \in (0, 1)$ is the household's bargaining power. Since meetings are anonymous, the cash constraint binds in equilibrium. The solution to (NB-h) yields the payment function

$$(2) \quad g(q) = (1 - \eta_h)v(q) + \eta_h q,$$

with $d = g(q) = z$ and household surplus $\eta_h[v(q) - q] \geq 0$.

The household's AD portfolio choice z' determines the DM₂ trade quantity through $g(q) = z'$. The marginal cost of holding one additional unit of real balances is the opportunity cost i ; the marginal benefit is the expected DM₂ surplus from a bilateral match. Setting the two equal, the first-order condition for z' is

$$(MD) \quad i = \bar{\alpha}_h \left[\frac{v'(q)}{g'(q)} - 1 \right],$$

which is the household's money demand schedule: $z'(i) = g(q^*(i))$ is strictly decreasing in i , as higher inflation makes real balances more costly to hold.² Because $i > 0$ taxes

²Figures report the annualized rate $i_{\text{ann}} = 4s$ on the horizontal axis.

money holdings, equilibrium q lies below the first-best level q^{FB} (attained at $i = 0$), which reduces the revenue each firm earns from decentralized trade. In Section 3, equation (MD) is combined with market clearing to yield the LW curve.

In the MP market, the household's continuation values satisfy

$$\begin{aligned} U_h^1(z) &= V_h^1(z) + \delta_L [V_h^0(z) - V_h^1(z)], \\ U_h^0(z) &= V_h^0(z) + \lambda_h [V_h^1(z) - V_h^0(z)], \end{aligned}$$

where δ_L is the exogenous separation rate and $\lambda_h = \mathcal{M}(u, v)/u$ is the job-finding rate. Wages are set by Nash bargaining with worker power η_L ; the solution is stated in Section 3.

2.3. Firms

A continuum of firms enters freely. In Stage 1, each firm posts one vacancy at flow cost c_v to recruit a worker. The labor matching function is $\mathcal{M}(u, v) = Zu^{1-\sigma_m}v^{\sigma_m}$; market tightness is $\tau \equiv v/u$. In steady state, $Z\tau^{\sigma_m} = \delta_L(1-u)/u$ pins down $\tau^*(u)$, and the job-filling rate is $\lambda_f(u) = Z(\tau^*(u))^{\sigma_m-1}$. Firms are heterogeneous only in their access to bank credit: *unbanked* firms finance Stage 3 capital entirely from money holdings; *banked* firms supplement cash with bank credit. Since workers cannot observe a firm's banking status, Nash wage bargaining with worker power η_L applies symmetrically to both types.

2.3.1. Unbanked Firms

In the AD market, an unbanked firm chooses real money holdings z'_f (equivalently, next-period capital $k_u = z'_f$) to maximize revenue net of the opportunity cost of holding money:

$$(AD-f) \quad \max_{z'_f \geq 0} \{F(z'_f) - (1+s)z'_f\},$$

where s is the nominal spread and $F(k) = A_F k^{1-\xi_F}/(1-\xi_F)$. The first-order condition $F'(k_u) = 1+s$ yields the demand for capital:

$$(3) \quad k_u(s) = \left(\frac{A_F}{1+s} \right)^{1/\xi_F}.$$

Higher inflation raises the opportunity cost of holding cash, so $k_u(s)$ is strictly decreasing in s : inflation depresses both households' money demand (reducing q) and unbanked firm investment simultaneously.

In Stage 4, each employed firm meets a household buyer with probability $\alpha_f(u)$ and earns expected DM₂ revenue $R^{\text{DM}_2}(u, q) \equiv \alpha_f(u)(1-\eta_h)[v(q) - q]$. Total period

net revenue—production output net of the inflation tax on money holdings—is

$$R_u(s, u, q) \equiv F(k_u(s)) - s k_u(s) + R^{\text{DM}_2}(u, q).$$

The term $-s k_u(s)$ is the opportunity cost of holding k_u units of real money balances at nominal spread s .

A matched firm retains its worker with probability $1 - \delta_L$ and separates with probability δ_L each period; free entry drives the vacancy value to zero. The steady-state value of a filled unbanked job therefore satisfies

$$J_u = R_u - w + \beta(1 - \delta_L) J_u,$$

or equivalently $(r + \delta_L)J_u = R_u - w$. Nash bargaining with worker power η_L gives $w = \eta_L R_u + (1 - \eta_L)b$.³ So

$$(4) \quad J_u(s, u, q) = \frac{(1 - \eta_L)(R_u(s, u, q) - b)}{r + \delta_L}.$$

2.3.2. Banked Firms: Optimal Capital Contract

A banked firm holds cash m_b and bank credit d_b , investing $k_b = m_b + d_b$. The bank bears monitoring cost $\psi(d_b) = \frac{B}{1+\gamma} d_b^{1+\gamma}$ ($B, \gamma > 0$). The firm's outside option is the unbanked value, which carries inflation tax $s k_u$ on its full cash holding. Nash bargaining between firm and bank determines d_b :

$$(NB-b) \quad \max_{d_b \geq 0} [F(k_b) - F(k_u(s)) + s(k_u(s) - m_b)]^{\eta_b} [s d_b - \psi(d_b)]^{1-\eta_b}.$$

The firm's surplus $F(k_b) - F(k_u) + s(k_u - m_b)$ is the gain in net revenue from banking: the banked firm earns $F(k_b) - s m_b$ while the unbanked alternative earns $F(k_u) - s k_u$, so the net gain is $[F(k_b) - s m_b] - [F(k_u) - s k_u]$. The bank's surplus $s d_b - \psi(d_b)$ is the spread income net of monitoring cost. In competitive credit markets at spread s , the bank maximises its own surplus by extending credit to the point $\psi'(d_b^*) = s$, which determines d_b^* independently of the split; Nash bargaining then allocates the total surplus between the two parties.

LEMMA 1 (Optimal capital contract). Define $\hat{s}$ by $k_u(\hat{s}) = (\hat{s}/B)^{1/\gamma}$. The Nash solution to (NB-b) has two regimes:

- (i) Low-spread regime ($s \leq \hat{s}$): $d_b^* = (s/B)^{1/\gamma}$, $m_b = k_u(s) - d_b^* > 0$, $k_b = k_u(s)$. The banked firm substitutes credit for cash but holds the same total capital as an unbanked firm.

³This is the standard intra-period Nash wage: each period, the match surplus $R_u - b$ is split in proportion $(\eta_L, 1 - \eta_L)$. Because workers cannot observe the firm's banking relationship, the same bargaining protocol applies to both types; the banked-firm wage is $w_b = \eta_L R_b + (1 - \eta_L)b > w_u$ whenever $R_b > R_u$.

- (ii) High-spread regime ($s > \hat{s}$): $d_b^* = (s/B)^{1/\gamma}$, $m_b = 0$, $k_b = (s/B)^{1/\gamma} > k_u(s)$. The banked firm is unconstrained and invests strictly more capital.

Proof. See Appendix A. $\square$

The two-regime structure reflects the interaction between the cash constraint $m_b \geq 0$ and the credit line. In the low-spread regime, the constraint is slack: the firm replaces costly cash with cheap credit one-for-one, leaving total capital $k_b = k_u(s)$ unchanged. The gain from banking is purely a saving on cash opportunity costs. Once s crosses $\hat{s}$, the firm exhausts its cash and the credit line continues to expand with s ; banked firms now invest more capital than unbanked firms, generating a persistent output advantage that grows with the spread. The threshold $\hat{s}$ therefore marks the point at which the credit relationship becomes a source of productive investment rather than merely a cash-management tool. The banked firm's total period *net* revenue—production output net of the inflation tax on its cash holdings m_b —is

$$R_b(s, u, q) \equiv F(k_b(s)) - s m_b(s) + R^{\text{DM}_2}(u, q),$$

where $m_b(s) = \max(k_u(s) - d_b^*(s), 0)$ is the firm's optimal cash holding from Lemma 1. By the same Bellman argument as (4):

$$(5) \quad J_b(s, u, q) = \frac{(1 - \eta_L)(R_b(s, u, q) - b)}{r + \delta_L}.$$

2.3.3. Surplus from a Banking Relationship

The surplus gain from banking is $\Delta_{\text{firm}}(s) \equiv J_b(s) - J_u(s)$:

$$(6) \quad \Delta_{\text{firm}}(s) = \delta_{\text{innov}} + \frac{(1 - \eta_L) \Pi(s)}{r + \delta_L},$$

where $\delta_{\text{innov}} > 0$ is a spread-independent innovation premium (Bethune et al. 2021) and

$$(7) \quad \Pi(s) = [F(k_b(s)) - F(k_u(s))] + s(k_u(s) - m_b^*(s)).$$

Here $m_b^*(s) \equiv \max(k_u(s) - d_b^*(s), 0)$ is the firm's optimal cash holding. The two terms have a clear economic interpretation: the first is the output gain from a larger capital stock (positive only in the high-spread regime $s > \hat{s}$, where $k_b > k_u$); the second, $s(k_u - m_b^*)$, is the *cash-saving* gain—the reduction in the inflation tax from replacing high-cost cash with bank credit.

In the low-spread regime ($s \leq \hat{s}$), $k_b = k_u$ and $k_u - m_b^* = d_b^*(s)$, so $\Pi(s) = s d_b^*(s)$: the entire surplus comes from cash saving. In the high-spread regime ($s > \hat{s}$), $m_b^* = 0$ and $\Pi(s) = [F(k_b) - F(k_u)] + s k_u(s)$: both capital investment and residual cash saving contribute.

LEMMA 2 (Monotone relationship surplus). $\Delta_{\text{firm}}(s)$ is strictly increasing in s for all $s \geq 0$.
Proof. See Appendix A. $\square$

The monotonicity holds in both regimes. In the low-spread regime, the cash-saving term $s d_b^*(s) = s(s/B)^{1/\gamma}$ rises with s both directly and through the expanding credit line $d_b^*(s)$. In the high-spread regime, the output-gain term $F(k_b) - F(k_u)$ additionally widens, since banked firms invest more capital while unbanked firms invest less. Consequently, higher inflation makes banking relationships more valuable to firms, attracting bank entry and increasing the banking fraction ℓ^* (Section 2.4). A larger ℓ^* lifts average net revenue $\bar{R}$ and hence average firm value $\bar{J}$, inducing firms to post more vacancies and reducing unemployment—a force that acts in the *opposite* direction to the inflation tax on household money demand.

Average firm *net* revenue, weighted by the banking fraction ℓ^* , subtracts the inflation tax on each type's capital:

$$(8) \quad \bar{R}(s, u, q, \ell^*) \equiv \ell^* [F(k_b(s)) - s m_b(s)] + (1 - \ell^*) [F(k_u(s)) - s k_u(s)] + \alpha_f(u)(1 - \eta_h)[v(q) - q].$$

so average firm value is $\bar{J} = (1 - \eta_L)(\bar{R} - b)/(r + \delta_L)$. The free-entry condition $c_v = \beta \lambda_f(u) \bar{J}$ is stated as an equilibrium condition in Section 3.

2.4. Banks

Banks enter the credit market freely, paying per-period entry cost $\kappa > 0$. In Stage 2, with $\theta_b \equiv v_b/n_b$ credit-market tightness, an unbanked firm meets a bank with probability $\alpha_b(\theta_b) = \bar{\alpha} \theta_b / (1 + \theta_b)$, and a bank meets an unbanked firm with probability $q_b(\theta_b) \equiv \bar{\alpha} / (1 + \theta_b)$. Existing relationships dissolve at rate δ_b per period. A bank's payoff from a relationship combines credit revenue and a non-credit flow ϕ_b from the innovation channel (technology transfer, advisory services; Bethune et al. 2021). The bank receives the spread s on credit d_b^* , incurs monitoring cost $\psi(d_b^*)$, and earns ϕ_b per period; its value when matched satisfies the Bellman equation

$$V_b^1 = s d_b^*(s) - \psi(d_b^*(s)) + \phi_b + \beta(1 - \delta_b) V_b^1,$$

or equivalently

$$(r + \delta_b) V_b^1 = s d_b^*(s) - \psi(d_b^*(s)) + \phi_b.$$

From the NB-b Nash solution in Section 2.3.2, the bank's share of the total match surplus equals $(1 - \eta_b) \Delta_{\text{firm}}(s)$, giving

$$(9) \quad V_b^1(s) = (1 - \eta_b) \Delta_{\text{firm}}(s).$$

Bellman–Nash consistency requires $(1 - \eta_b) \Delta_{\text{firm}}(s) (r + \delta_b) = s d_b^*(s) - \psi(d_b^*(s)) + \phi_b$ for all s . At $s = 0$, $d_b^*(0) = 0$ and $\psi(0) = 0$, so $(1 - \eta_b) \delta_{\text{innov}} (r + \delta_b) = \phi_b$; this pins down ϕ_b (and hence δ_{innov}) so that $V_b^1(0) = (1 - \eta_b) \delta_{\text{innov}} > 0$, consistent with the Nash share at Friedman rule. For $s > 0$, the condition determines δ_{innov} jointly with the other calibration targets (Appendix B). V_b^1 is strictly increasing in s by Lemma 2: higher inflation raises the value of a bank–firm relationship, attracting bank entry.

An unmatched bank pays κ each period and meets an unbanked firm with probability $q_b(\theta_b)$. Its continuation value satisfies

$$V_b^0 = -\kappa + \beta [q_b(\theta_b) V_b^1 + (1 - q_b(\theta_b)) V_b^0].$$

Free entry drives the value of entering the credit market to zero, $V_b^0 = 0$. Substituting and using (9) yields the **bank free-entry condition (BFE)**:

$$(BFE) \quad \kappa = \beta q_b(\theta_b) (1 - \eta_b) \Delta_{\text{firm}}(s).$$

Solving for credit-market tightness:

$$(10) \quad \theta_b^*(s) = \frac{\beta (1 - \eta_b) \Delta_{\text{firm}}(s) \bar{\alpha}}{\kappa} - 1 \geq 0.$$

The steady-state banking fraction (fraction of *all* active firms holding a bank relationship) satisfies the flow balance $\alpha_b(\theta_b^*)(1 - \ell^*) = \delta_b \ell^*$:

$$(11) \quad \ell^*(s) = \frac{\alpha_b(\theta_b^*(s))}{\alpha_b(\theta_b^*(s)) + \delta_b},$$

where $1 - \ell^*$ is the unbanked share and the denominator $\alpha_b + \delta_b$ is the sum of the entry and destruction rates from the flow balance. Since $\Delta_{\text{firm}}(s)$ is strictly increasing (Lemma 2), θ_b^* and hence ℓ^* are also strictly increasing in s : higher inflation raises bank entry, increasing the fraction of banked firms. Importantly, equations (BFE)–(11) involve s alone, independently of (u, q) : the credit market can be solved first and its outcome $\ell^*(s)$ taken as given when solving the labor-market system.

2.5. Equilibrium

We now define the equilibrium concept and state the conditions that jointly determine the stationary equilibrium $(u^*, q^*, \ell^*, \theta_b^*)$.

DEFINITION 1 (Stationary Monetary Equilibrium). *A stationary monetary equilibrium given $i \geq 0$ is a tuple $(u^*, q^*, \ell^*, \theta_b^*)$ such that: (i) the bank free-entry condition (BFE) holds; (ii) the household portfolio condition (LW) holds; (iii) the firm free-entry condition (MP) holds.*

With meeting probabilities evaluated at $\bar{u}$, $\bar{\alpha}_h$ is a constant (Liu 2008). The household's first-order condition for real money balances is the **LW curve**:

$$(LW) \quad i = \bar{\alpha}_h \left[\frac{v'(q)}{g'(q)} - 1 \right].$$

Equating the marginal cost of holding money to its expected marginal benefit, this condition determines $q^*(i)$ independently of u .

The firm's free-entry condition for vacancies, $J^0 = 0$, is the **MP curve**:

$$(MP) \quad c_v = \frac{\beta \lambda_f(u) (1 - \eta_L) (\bar{R} - b)}{r + \delta_L}.$$

A higher $\bar{R}$ —driven by more banked firms or higher trade volume—raises $\bar{J}$, inducing vacancy posting until the cost c_v is exhausted.

The Liu (2008) device guarantees a unique stationary monetary equilibrium. In a standard Lagos–Wright framework with endogenous matching, the household's DM contact probability $\alpha_h(u)$ depends on the labor market: higher unemployment lowers α_h , shifts money demand, and alters equilibrium DM trade q^* . This two-way feedback between (q^*, u^*) can generate equilibrium multiplicity. By evaluating the contact probability at its steady-state value $\bar{\alpha}_h$, as in Liu (2008), the LW curve (LW) becomes a mapping from the nominal rate i to q^* alone, independently of u . The three equilibrium conditions then form a sequential system: the credit market is solved first for (θ_b^*, ℓ^*) , then the goods market for q^* , and finally the labor market for u^* . Each step involves a strictly monotone mapping and yields a unique solution.

PROPOSITION 1 (Uniqueness). *For any $i \geq 0$, a unique stationary monetary equilibrium exists.*

PROOF. See Appendix A. □

3. Two Channels and the Hump-Shaped Locus

With the equilibrium defined in Section 2.5, we now analyze how $u^*(i)$ responds to changes in i . Section 3.1 decomposes the match surplus into two competing channels and derives the conditions under which their interaction produces a hump-shaped locus. Section 3.2 presents the quantitative properties under the benchmark calibration.

3.1. The liquidity channel vs. the financial intermediation channel

The match surplus $J^*(i)$ decomposes additively into two components:

$$(12) \quad J^*(i) = \underbrace{\frac{(1 - \eta_L)[F(k_u^*) - s k_u^* + \bar{\alpha}_f(1 - \eta_h)(v(q^*) - q^*) - b]}{r + \delta_L}}_{J_{\text{base}}(i)} + \underbrace{\frac{(1 - \eta_L) \ell^*(i) [F(k_b^*) - F(k_u^*) + s(k_u^* - m_b^*)]}{r + \delta_L}}_{J_{\text{bank}}(i)},$$

where J_{base} is the counterfactual match value in a pure BMW economy without bank credit ($\ell^* \equiv 0$), and J_{bank} is the additional surplus that banking relationships contribute. Since J^* governs the job-creation condition (MP), any force that raises J^* lowers equilibrium unemployment, and vice versa.

liquidity channel. Higher inflation erodes firms' DM revenue by compressing workers' real money balances. Households carry money across periods to finance DM purchases at a cost of i per unit held; as inflation rises, buyers economize on real balances and the equilibrium quantity q^* falls, as required by the LW condition (LW). Each firm's DM revenue, $\bar{\alpha}_f(1 - \eta_h)[v(q^*) - q^*]$, falls more than proportionately because v is strictly concave, so even a moderate compression of q^* disproportionately reduces the firm's bilateral surplus.

Unbanked firms also bear the full inflation tax on capital, reducing productive investment and compressing output per match. A firm without a banking relationship must carry its investment funds as cash; its capital optimality condition, $F'(k_u^*) = 1 + s$, implies k_u^* falls strictly as s rises (strict concavity of F), compressing output per match through $F(k_u^*)$. Both margins reduce the present value of a filled position $J^*(i)$, and the MP condition (MP) then requires lower market tightness $\lambda_f(u^*)$, so steady-state unemployment rises. Neither margin contains an internal offsetting force: the *liquidity channel* operates without interruption from the Friedman rule upward.

financial intermediation channel. Bank credit raises the value of a job match by severing the link between inflation and the cost of capital. Without a banking relationship, a firm must carry its investment funds as cash through the production period, paying the spread s per unit held; its capital optimality condition, $F'(k_u^*) = 1 + s$, forces the equilibrium capital stock down continuously as inflation rises. A banking relationship severs this link: the bank provides credit that is repaid after production, so investment funds need not

be held as cash and the firm bears no inflation tax on capital. The banked firm therefore invests at a level k_b^* strictly above $k_u^*(s)$, produces more output per match, and earns a higher match value $J_b > J_u$. The real effect of the *financial intermediation channel* is not a financial saving on cash balances but a restoration of productive capital — and through it, a higher incentive to post vacancies and lower unemployment.

Above the regime threshold $\hat{s}$, a banked firm invests strictly more than an unbanked firm, and the capital gap widens with inflation. An unbanked firm solves $\max_k F(k) - (1+s)k$, equating the marginal product of capital to the carrying cost: $F'(k_u^*) = 1+s$. By strict concavity of F , this implies $dk_u^*/ds < 0$ — the unbanked firm's capital stock falls continuously with the spread. In the low-spread regime ($s \leq \hat{s}$), the banked firm's credit line does not yet exceed its cash holdings, so $k_b^*(s) = k_u^*(s)$ (Lemma 1); the benefit of banking is purely a saving on the cash opportunity cost. In the high-spread regime ($s > \hat{s}$), $k_b^*(s) = (s/B)^{1/\gamma}$ grows while $k_u^*(s)$ falls, so the differential $k_b^* - k_u^*$ widens strictly with inflation. Because F is strictly increasing, the output gap $F(k_b^*) - F(k_u^*)$ widens as well: every unit of inflation drives a growing wedge between the productivity of a banked match and that of an unbanked match.

The net gain from bank credit, $\Delta_{\text{firm}}(s) = \delta_{\text{innov}} + (1-\eta_L)\Pi(s)/(r+\delta_L)$ (equation (6)), is strictly increasing in s for all $s \geq 0$ (Lemma 2). At the Friedman rule $s = 0$, both components of Π vanish ($k_b^*(0) = k_u^*(0)$ and $d_b^*(0) = 0$), leaving $\Delta_{\text{firm}}(0) = \delta_{\text{innov}} > 0$. The innovation premium $\delta_{\text{innov}} > 0$ ensures that banking relationships are individually rational even at zero inflation: the benchmark calibration (Appendix B) sets δ_{innov} to match the Friedman-rule banking fraction, confirming that a majority of firms maintain banking relationships even at zero inflation.

The regime threshold $\hat{s}$ determines when the *capital advantage* of banking activates, not when banking begins. Below $\hat{s}$, the credit line $(s/B)^{1/\gamma}$ falls short of the firm's optimal cash holdings, so the banked firm substitutes credit for cash one-for-one with no change in total capital ($k_b^* = k_u^*$, Lemma 1); the gain from banking is purely a saving on the inflation tax on cash. At $\hat{s}$, the credit line exactly matches the unbanked firm's optimal capital $k_u^*(\hat{s})$: the bank fully displaces the firm's self-finance; above $\hat{s}$, the credit line exceeds k_u^* , and the banked firm holds strictly more capital than the unbanked firm. The position of $\hat{i}$ is determined by the monitoring-cost scale B : a higher B demands a larger spread before the credit line exceeds the self-financing optimum, pushing $\hat{i}$ upward and shifting the unemployment peak toward higher inflation.

LEMMA 3 (Regime threshold). *There exists a unique threshold $\hat{s} > 0$, defined by $k_u(\hat{s}) = (\hat{s}/B)^{1/\gamma}$, at which the capital channel activates.*

- (i) $k_b^*(s) = k_u^*(s)$ for all $s \leq \hat{s}$: banked and unbanked firms invest identically; the financial intermediation channel gain is purely a saving on the cash opportunity cost.
- (ii) $k_b^*(s) > k_u^*(s)$ for all $s > \hat{s}$: the capital-investment advantage operates.

- (iii) $\ell^*(s) > 0$ for all $s \geq 0$, and $d\ell^*/ds > 0$: banking is profitable from the Friedman rule and the banking fraction rises continuously with inflation.

Proof. See Appendix A. □

As inflation rises above $\hat{i}$, the *financial intermediation channel* grows strictly stronger, in direct contrast to the *liquidity channel*. The gain $\Delta_{\text{firm}}(s)$ increases with s (Lemma 2), so at higher spreads a greater fraction of firms finds bank credit individually rational; the extensive margin — more firms choosing to bank — amplifies the intensive margin — each banked firm investing more than its unbanked counterpart. The two margins reinforce each other: a higher s simultaneously widens the output gap $F(k_b^*) - F(k_u^*)$ per banked match and raises the share ℓ^* of matches that benefit from it, so the average match value rises at an accelerating rate with inflation.

PROPOSITION 2 (Channel decomposition). *The steady-state effects of inflation decompose into two channels:*

- (i) $dq^*/di < 0$ for all $i > 0$: higher inflation compresses the equilibrium DM quantity.
- (ii) $A_{\text{Liq}} > 0$ for all $i > 0$: the liquidity channel raises unemployment monotonically with inflation.
- (iii) $d\ell^*/di > 0$ for all $i > 0$: the banking fraction rises strictly from the Friedman rule.
- (iv) $A_{\text{FI}} = 0$ for $i \leq \hat{i}$ and $A_{\text{FI}} < 0$ for $i > \hat{i}$: the financial intermediation channel is dormant below $\hat{i}$ and lowers unemployment above it.⁴

PROOF. See Appendix A. □

Figure 1 illustrates the two forces quantitatively. Panel (a) shows that J_{base} declines monotonically from the Friedman rule, driven initially by the capital-cost margin $F(k_u^*) - s k_u^*$ and increasingly by the DM-surplus compression at higher inflation. Panel (b) shows that J_{bank} is negligible below $\hat{i}$ (cash-saving only) and then rises sharply above $\hat{i}$ as the capital-investment margin activates. The net match value therefore falls below $\hat{i}$ (liquidity-dominated) and reverses above it (credit-dominated), producing the hump-shaped unemployment locus.

The capital regime transition underlying panel (b) is shown in Figure 2. Below $\hat{i}$, banked and unbanked firms hold identical capital; above $\hat{i}$, the capital gap $k_b^* - k_u^*$ widens strictly with inflation as the credit line exceeds the self-financing optimum.

The banking fraction $\ell^*(i)$ makes the kink structure concrete (Figure 3). DM trade $q^*(i)$ falls smoothly throughout with no kink at $\hat{i}$, while ℓ^* rises slowly below $\hat{i}$ and then steeply above it as the capital premium activates.

⁴The decomposition assigns the cash-saving margin of the *financial intermediation channel*—the reduction in capital-financing costs for banked firms—to A_{Liq} rather than A_{FI} ; this is why the hump peak $i^\dagger$ lies slightly below $\hat{i}$ (see Section 3.2).

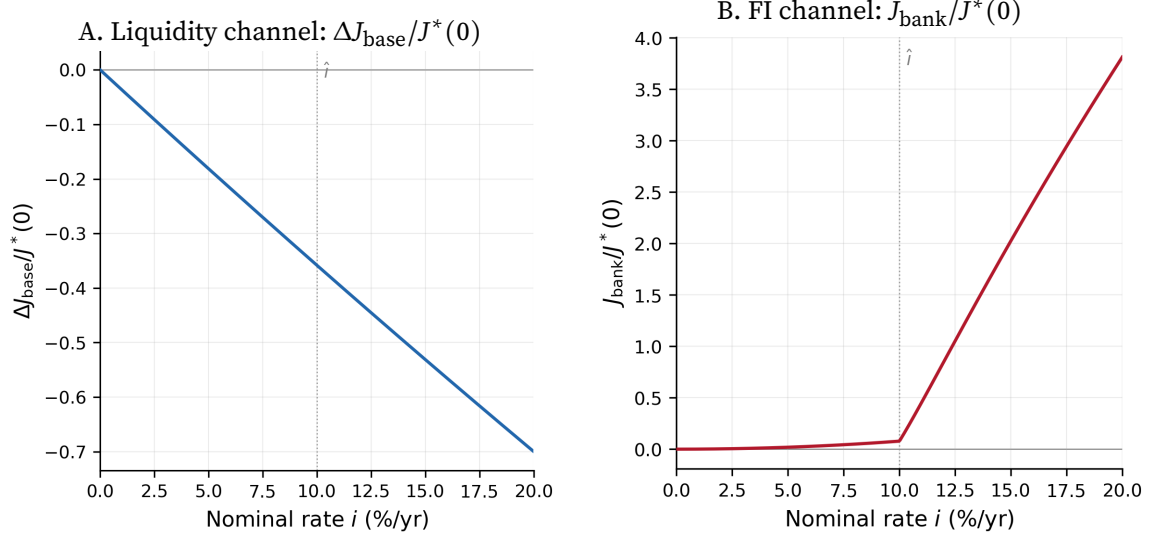


FIGURE 1. Two opposing forces on the match surplus.

3.2. The Hump-Shaped Unemployment Locus

Figure 4 shows the equilibrium unemployment locus $u^*(i)$ under the benchmark calibration. Starting from $u^*(0) = 5\%$ at the Friedman rule, unemployment rises gradually as inflation increases, peaks near the regime threshold $\hat{i} = 10\%/yr$, and then reverses sharply as inflation rises further. The pre-peak rise is modest; the post-peak decline is substantially larger—the locus is markedly asymmetric around the peak.

Below the regime threshold, the financial intermediation channel provides no capital premium: by Lemma 1, $k_b^* = k_u^*$ throughout this range, so $J_{\text{bank}} = 0$ identically. Banking relationships are widespread—firms already benefit from the cash-saving component of bank credit—but the capital advantage of banking has not yet activated. The entire effect of inflation therefore operates through J_{base} : rising s erodes workers' real balances, compresses DM trade q^* , and reduces the DM surplus that firms capture through Nash bargaining. With no countervailing force from the credit market, unemployment rises monotonically and continuously with inflation.

Above $\hat{i}$, the capital advantage of banking switches on and two self-reinforcing margins operate simultaneously. On the *extensive margin*, higher inflation raises the value of banking relationships $\Delta_{\text{firm}}(s)$, attracting bank entry and increasing the fraction of banked firms ℓ^* . On the *intensive margin*, banked and unbanked firms' capital choices diverge: k_b^* rises with s while k_u^* continues to fall (Figure 2). Both margins amplify J_{bank} , which grows steeply above $\hat{i}$, more than offsetting the continued erosion of J_{base} . The net match surplus J^* therefore rises, vacancy creation recovers, and unemployment declines.

The locus is structurally asymmetric because the two channels differ in their activation dynamics. The liquidity channel is continuous and attenuating: it compresses DM trade

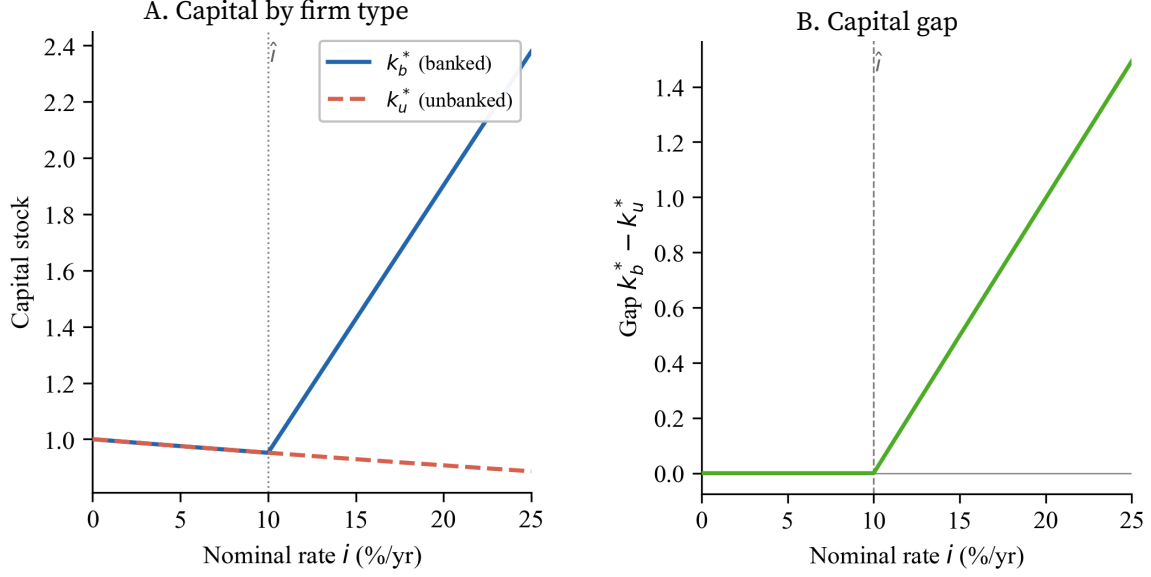


FIGURE 2. Capital regime transition.

the moment inflation departs from the Friedman rule, but its marginal effect diminishes as the DM surplus shrinks. The financial intermediation channel is discrete and amplifying: dormant until $\hat{i}$, it then activates both margins simultaneously, generating superlinear growth in J_{bank} as the capital gap $k_b^* - k_u^*$ widens (Figure 2, right panel). The asymmetry has a direct policy implication: disinflation from above $\hat{i}$ unwinds this channel and produces a substantially larger employment loss than an equal inflation increase from the Friedman rule.

4. Fragility and Financial Development

The baseline model treats the relationship destruction rate δ_b as a constant and the monitoring cost B as a fixed parameter. This section relaxes each assumption in turn. Section 4.1 shows that endogenous relationship fragility generates an N-shaped unemployment locus, reinforcing the limits of monetary expansion. Section 4.2 shows that financial development—a lower monitoring cost B —shifts the non-monotone locus leftward, generating testable cross-country predictions.

4.1. Endogenous Relationship Fragility

Endogenous fragility disciplines the financial intermediation channel and can generate a third, rising arm in the unemployment locus. Suppose that inflation weakens lending

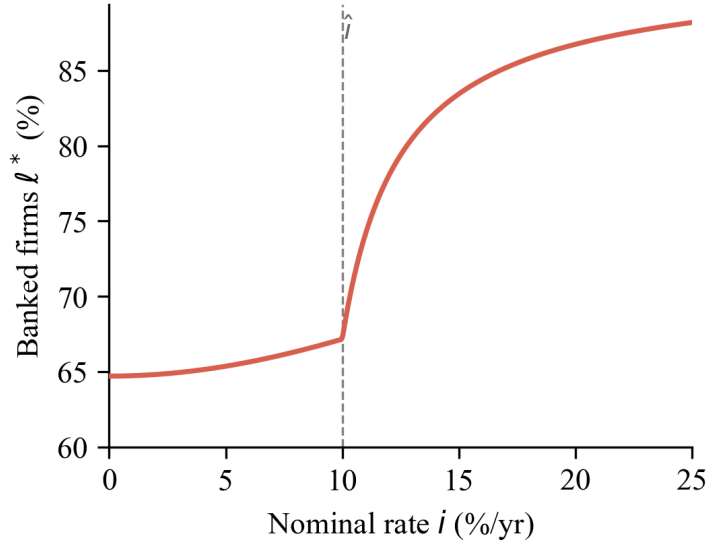


FIGURE 3. Banking fraction.

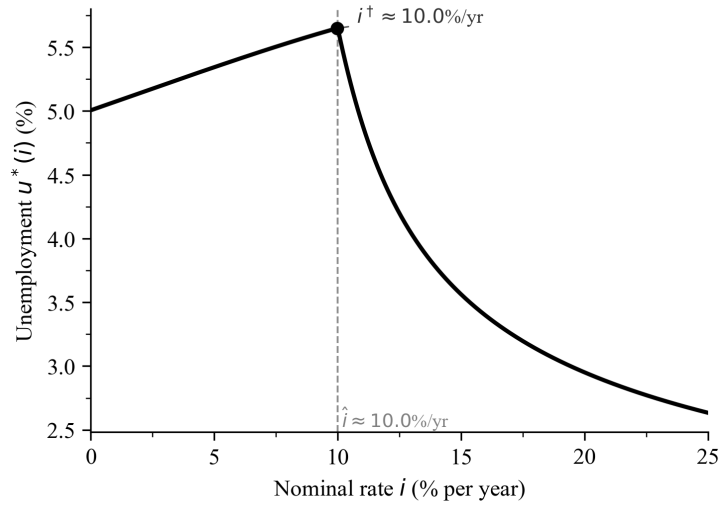


FIGURE 4. Inflation-unemployment locus.

relationships at rate $\varphi \geq 0$:

$$(F1) \quad \delta_b(s) = \delta_{b0} + \varphi s.$$

Nominal volatility makes credit monitoring more costly and inflation-driven portfolio rebalancing accelerates counterparty turnover, so $\varphi > 0$ is empirically plausible (Boyd et al. 2001).

Fragility introduces a tension between two opposing forces. Higher inflation raises the firm-bank surplus $\Delta_{\text{firm}}(s)$, attracting bank entry—the *entry effect*. At the same time, a

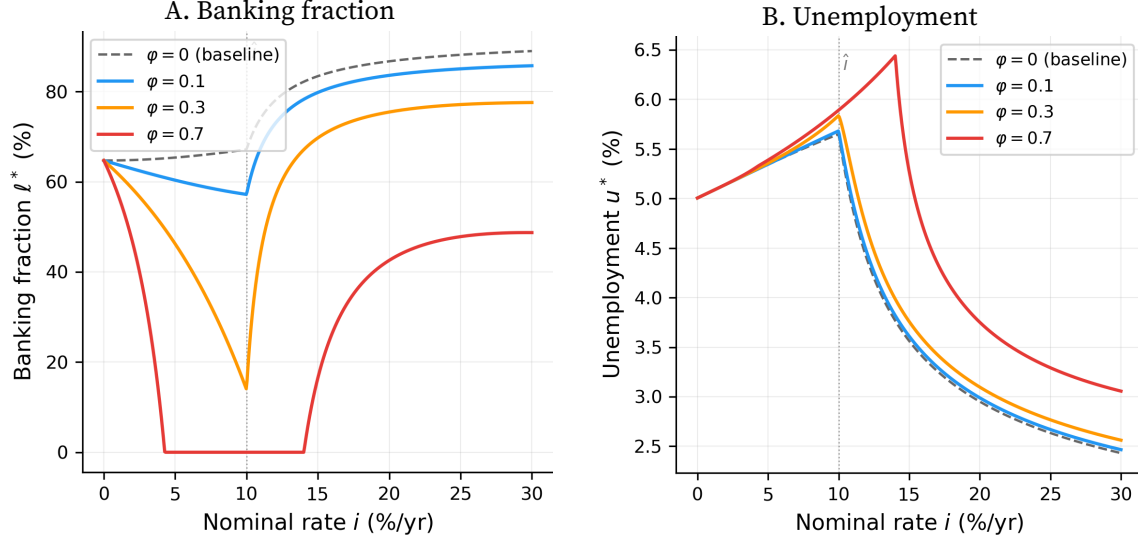


FIGURE 5. Endogenous fragility.

higher spread shortens the expected duration of each lending relationship, reducing the bank's continuation value and discouraging entry—the *fragility effect*. The banking fraction rises when the entry effect dominates and falls when fragility dominates. At low inflation the entry effect prevails; sufficiently high φ eventually reverses the balance, causing ℓ^* to peak and then decline.

With mild fragility the non-monotone unemployment locus is preserved, though with a somewhat attenuated post-peak employment gain, as the banking fraction rises more slowly than in the baseline. With strong fragility, the destruction rate outpaces matching at moderate inflation: the banking fraction collapses, recovers only when the capital-investment margin becomes large enough to compensate, and is subsequently bounded well below the baseline (Figure 5). The unemployment peak shifts to higher inflation and the post-peak decline shrinks. At extreme fragility, banking collapses entirely at high inflation and unemployment develops a third rising arm—a genuine N-shaped locus.

Endogenous fragility provides a natural limit on the employment gains from inflation. In the baseline ($\varphi = 0$), the financial intermediation channel grows without bound as ℓ^* approaches its ceiling; there is no endogenous force that arrests the unemployment decline. With $\varphi > 0$, the very mechanism that creates banking relationships—a high nominal spread—simultaneously destroys them. The resulting tension imposes an endogenous ceiling on ℓ^* and hence on the employment gain: a policy of sustained high inflation to exploit the non-monotone locus is self-limiting, with the financial system's resilience to inflation determining how far down the post-peak arm the economy can travel.

4.2. Financial Development and the Phillips Curve

Financially developed economies activate the Financial Intermediation Channel at lower inflation, shifting the non-monotone locus leftward along the inflation axis. Consider economies that differ in the monitoring cost B : a lower B corresponds to cheaper credit monitoring, making banking relationships more profitable at any given spread.

Financial development operates exclusively through the regime threshold $\hat{s}$, defined by $k_u(\hat{s}) = (\hat{s}/B)^{1/\gamma}$. Reducing B lowers $\hat{s}$, activating the capital channel at a lower nominal rate: the capital premium $k_b^* - k_u^*$ begins to emerge at milder inflation, and the banked capital schedule $k_b^*(s) = (s/B)^{1/\gamma}$ is higher at every spread above $\hat{s}(B)$. Both effects reinforce the credit channel. Crucially, B does not affect DM trade: the LW equation pins down $q^*(i)$ through household preferences and the nominal rate alone, independently of monitoring costs. The goods-market cost of inflation—the decline in q^* and the associated DM-surplus loss—is therefore identical across financial structures. Financial development improves employment outcomes without touching the *liquidity channel*.

The banking fraction is identical across financial structures at the Friedman rule: with zero nominal spread the capital premium vanishes and monitoring costs are irrelevant, leaving the banking fraction determined by the innovation channel alone. As inflation rises, differences in activation timing drive the curves apart. The gap is largest at moderate inflation, where some economies have already crossed their threshold and others have not; at high inflation, bank matching saturates and all curves converge.

Unemployment responds with a corresponding horizontal shift. A lower B compresses the hump peak slightly, because the goods-market channel has fewer periods of unopposed operation before the credit channel intervenes. Beyond the peak, the amplitude of the post-peak employment gain is broadly similar across financial structures: financial development shifts the non-monotone locus *horizontally*—to lower inflation—without dramatically altering its depth (Figure 6).

Financially deeper economies thus exhibit a more favorable inflation–unemployment tradeoff: the Phillips curve non-monotonicity appears at milder inflation and the economy enters the employment-rich post-peak region sooner. This extension is complementary to the fragility result in Section 4.1: financial development (B) determines *when* the credit channel activates—the location of $\hat{i}$ —while fragility (φ) determines *how far* the channel can push unemployment down, setting the ceiling on employment gains.

5. Conclusion

Introducing endogenous bank–firm credit relationships into a monetary search model fundamentally alters the standard prediction that inflation raises unemployment monotonically. Inflation operates through two opposing forces: a goods-market channel that

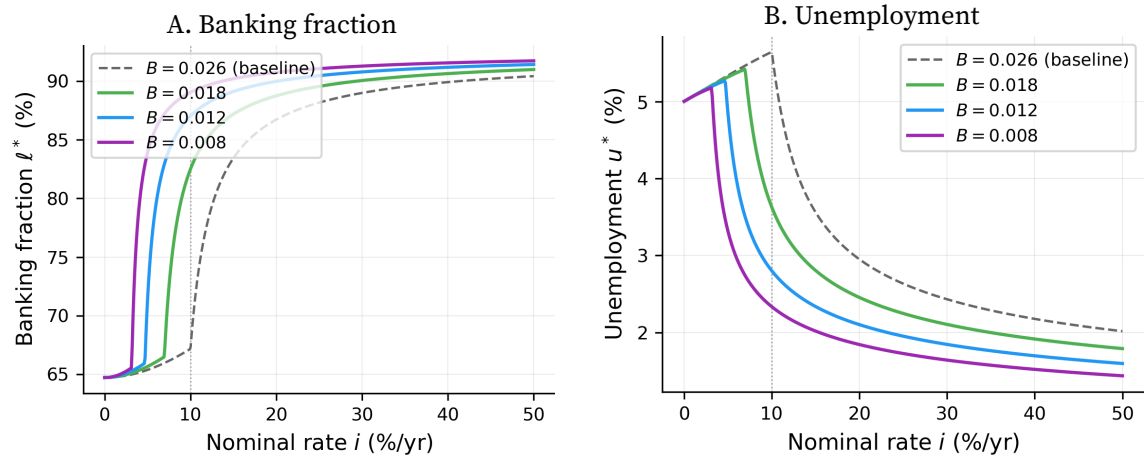


FIGURE 6. Financial development.

erodes real money balances and discourages vacancy creation, and a credit channel that, above an endogenous regime threshold, raises the return to lending relationships, drawing firms into banking and expanding their capital investment. The calibrated model produces a hump-shaped locus in which unemployment rises to a peak near ten percent annual inflation and then reverses sharply—with a post-peak decline roughly five times the pre-peak rise—as credit-market effects come to dominate.

Two limitations are worth noting. The analysis is confined to steady states: the regime threshold $\hat{i}$ is treated as invariant, whereas in practice credit-market conditions shift over the business cycle and may move the location of the hump endogenously. In addition, the model abstracts from the welfare costs of banking relationships—monitoring costs, entry costs, and potential capital misallocation—so the employment gains from inflation do not translate directly into welfare gains; the full welfare implications of the non-monotone locus remain uncharacterized.

Several directions remain open. Embedding aggregate shocks would allow the model to address state-dependent inflation effects and time-varying Phillips curves, with the threshold $\hat{i}$ shifting with credit conditions. Cross-country data on banking penetration and inflation histories could provide a quantitative test of the prediction that financially deeper economies exhibit a lower-inflation hump. Finally, characterizing optimal monetary policy in this environment—trading the inflation tax on liquidity against the credit-market benefit—would sharpen the policy implications of the non-monotone locus.

A. Proofs

Proof of Proposition 1

Step 1 (Bank market). Given nominal spread s , the right-hand side of (BFE), $\beta q_b(\theta_b)(1 - \eta_b)\Delta_{\text{firm}}(s)$, is strictly decreasing in θ_b from $\beta \bar{\alpha}(1 - \eta_b)\Delta_{\text{firm}}(s)$ (at $\theta_b = 0$) to 0 (as $\theta_b \rightarrow \infty$). Hence (BFE) has a unique solution $\theta_b^*(s) \geq 0$ whenever $\kappa \leq \beta \bar{\alpha}(1 - \eta_b)\Delta_{\text{firm}}(s)$, and $\theta_b^* = 0$ otherwise. The share $\ell^*(s)$ follows uniquely from (11).

Step 2 (DM goods market). The right-hand side of (LW), $\bar{\alpha}_h[v'(q)/g'(q) - 1]$, is strictly decreasing in q with range $(0, +\infty)$ as $q \in (0, q^{\text{FB}})$. Hence there is a unique $q^*(i) \in (0, q^{\text{FB}}]$ for each $i \geq 0$.

Step 3 (Labor market). Given (q^*, ℓ^*) , define the MP residual $f(u) \equiv \beta \lambda_f(u)(1 - \eta_L)(\bar{R} - b)/(r + \delta_L) - c_v$. Since $\lambda_f(u) = Z \tau(u)^{\sigma_m - 1}$ with $\tau(u) = [\delta_L(1 - u)/(Zu)]^{1/\sigma_m}$, and $\sigma_m < 1$ implies the exponent $(\sigma_m - 1)/\sigma_m < 0$, an increase in u lowers τ and therefore raises λ_f (the exponent on τ is negative). Hence f is strictly increasing in u . At the boundaries: as $u \rightarrow 0$, $\tau \rightarrow \infty$ so $\lambda_f = Z \tau^{\sigma_m - 1} \rightarrow 0$ and $f(0^+) = -c_v < 0$; as $u \rightarrow 1$, $\tau \rightarrow 0$ so $\lambda_f \rightarrow +\infty$ and $f(1^-) > 0$. The boundary conditions $f(0^+) < 0 < f(1^-)$ together with the strict monotonicity of f yield, by the intermediate-value theorem, a unique $u^* \in (0, 1)$. $\square$

Proof of Lemma 1

The matched firm-bank pair chooses credit d_b and cash m_b to maximize total match surplus, subject to $k_b = m_b + d_b$ and $m_b \geq 0$.

Step 1 (Credit level). Under transferable-utility Nash bargaining, the contract first maximises total match surplus and then distributes it. Total surplus from the credit relationship (net of monitoring) is $\Pi(s) - \psi(d_b)$, where $\Pi(s) = [F(k_b) - F(k_u)] + s(k_u - m_b)$ is the joint output and cash-saving gain. Because the firm's optimal cash holding m_b is determined in Step 2, the first-order condition with respect to d_b reduces to the bank's own marginal-profit condition $\psi'(d_b) = s$.⁵ The bank's monitoring cost $\psi(d_b) = \frac{B}{1+\gamma} d_b^{1+\gamma}$ is strictly convex with $\psi'(0) = 0$, so $\psi'(d_b^*) = B d_b^{*\gamma} = s$ yields the unique solution

$$d_b^*(s) = (s/B)^{1/\gamma} \quad \text{for all } s > 0, \quad d_b^*(0) = 0.$$

The Nash split of the resulting surplus $\Pi(s)$ is determined in Section 2.3.2.

Step 2 (Firm's investment). Given d_b^* , the firm chooses $m_b \geq 0$ to maximize $F(m_b + d_b^*) - (1 + s)m_b$, where $F'(k) = A_F k^{-\xi_F}$ is strictly decreasing. If $m_b > 0$ (interior), the first-order condition gives $F'(k_b) = 1 + s = F'(k_u)$, hence $k_b = k_u(s)$ by strict concavity, with $m_b = k_u(s) - d_b^*$.

Step 3 (Two regimes). The interior solution requires $m_b = k_u(s) - d_b^* \geq 0$.

- (i) *Low-spread regime* ($s \leq \hat{s}$): $k_u(s) \geq (s/B)^{1/\gamma}$, so $m_b > 0$, $k_b = k_u(s)$, and the banked firm holds the same total capital as an unbanked firm.
- (ii) *High-spread regime* ($s > \hat{s}$): $k_u(s) < (s/B)^{1/\gamma}$, so $m_b = 0$ and $k_b = d_b^* = (s/B)^{1/\gamma} > k_u(s)$.

Step 4 (Threshold). $k_u(s) = (A_F/(1+s))^{1/\xi_F}$ is continuous and strictly decreasing from A_F^{1/ξ_F} to 0; $(s/B)^{1/\gamma}$ is continuous and strictly increasing from 0 to ∞ . By the intermediate value theorem, the equation $k_u(\hat{s}) = (\hat{s}/B)^{1/\gamma}$ has a unique solution $\hat{s} > 0$. $\square$

Proof of Lemma 2

By (6), $\Delta_{\text{firm}}(s) = \delta_{\text{innov}} + (1 - \eta_L) \Pi(s)/(r + \delta_L)$, where $\delta_{\text{innov}} > 0$ is constant and, from (7),

$$\Pi(s) = [F(k_b(s)) - F(k_u(s))] + s(k_u(s) - m_b^*(s)), \quad m_b^*(s) \equiv \max(k_u(s) - d_b^*(s), 0).$$

It suffices to show $\Pi'(s) > 0$ for all $s > 0$.

Low-spread regime ($0 < s \leq \hat{s}$). $k_b = k_u$ and $m_b^* = k_u - d_b^*$, so $\Pi(s) = s d_b^*(s) = s (s/B)^{1/\gamma}$. Define $h(s) \equiv s (s/B)^{1/\gamma}$. Then $h'(s) = (1 + 1/\gamma) (s/B)^{1/\gamma} > 0$, so $\Pi'(s) = h'(s) > 0$.

⁵In the low-spread regime, total surplus $= s d_b - \psi(d_b)$ so $\partial/\partial d_b = s - \psi'(d_b) = 0$ gives $\psi'(d_b^*) = s$ directly. In the high-spread regime, the firm absorbs all credit into capital ($m_b = 0$), and the incremental firm gain from raising d_b is $F'(d_b) > s$ at the bank's optimum, so the binding constraint is the bank's profit condition. In both regimes the TU-Nash contract and the bank's individual profit-maximisation coincide at $\psi'(d_b^*) = s$.

High-spread regime ($s > \hat{s}$). $m_b^* = 0$, so $\Pi(s) = F(k_b) - F(k_u) + s k_u$. $k_b = (s/B)^{1/\gamma}$ and $k_u = (A_F/(1+s))^{1/\xi_F}$. Differentiating:

$$\Pi'(s) = \underbrace{F'(k_b) \frac{dk_b}{ds}}_{>0} - \underbrace{F'(k_u) \frac{dk_u}{ds}}_{<0} + \underbrace{k_u + s \frac{dk_u}{ds}}_{>0 \text{ (shown below)}}.$$

The first term is positive: $F'(k_b) > 0$ and $dk_b/ds > 0$. The second term contributes positively: $F'(k_u) > 0$ and $dk_u/ds < 0$. For the third term: $k_u + s k_u' = k_u - s k_u(1+s)^{-1} \xi_F^{-1}$ using $k_u' = -k_u \xi_F^{-1}/(1+s)$ (from $k_u = (A_F/(1+s))^{1/\xi_F}$). This equals $k_u[1 - s/(\xi_F(1+s))] > 0$ since $s/(\xi_F(1+s)) < 1$ for all finite s and $\xi_F > 0$. Hence $\Pi'(s) > 0$.

Continuity. At $s = \hat{s}$: $d_b^*(\hat{s}) = k_u(\hat{s})$ implies $m_b^*(\hat{s}) = 0$ and $F(k_b(\hat{s})) = F(k_u(\hat{s}))$. Both formulas give $\Pi(\hat{s}) = \hat{s} k_u(\hat{s})$. At $s = 0$: $\Pi(0) = 0$ (since $k_b(0) = k_u(0)$ and $d_b^*(0) = 0$). Combining with $\Pi'(s) > 0$ for all $s > 0$ completes the proof. $\square$

Proof of Lemma 3

Part (i)–(ii): Capital threshold. By Lemma 1, the threshold $\hat{s}$ defined by $k_u(\hat{s}) = (\hat{s}/B)^{1/\gamma}$ is unique and positive (existence and uniqueness follow because $k_u(s)$ is strictly decreasing with $k_u(0) > 0$ and $(s/B)^{1/\gamma}$ is strictly increasing from 0, guaranteeing a unique crossing). For $s \leq \hat{s}$, the Nash solution to the capital contract gives $k_b^*(s) = k_u^*(s)$; for $s > \hat{s}$, $k_b^*(s) = (s/B)^{1/\gamma} > k_u^*(s)$.

Part (iii): Banking fraction. At $s = 0$: by (6) and $\Pi(0) = 0$, $\Delta_{\text{firm}}(0) = \delta_{\text{innov}} > 0$. The BFE (BFE) with $\alpha_b(\theta_b) = \bar{\alpha}\theta_b/(1+\theta_b)$ yields $\theta_b^*(0) = \beta(1-\eta_b)\delta_{\text{innov}}\bar{\alpha}/\kappa - 1$. The calibration sets δ_{innov} so that $\beta(1-\eta_b)\delta_{\text{innov}}\bar{\alpha} > \kappa$, giving $\theta_b^*(0) > 0$ and hence $\ell^*(0) > 0$.

Monotonicity. By Lemma 2, $\Delta_{\text{firm}}(s)$ is strictly increasing in s . Hence $\theta_b^*(s)$ is strictly increasing in s via the BFE, and $\ell^*(s) = \bar{\alpha}\theta_b^*/(\bar{\alpha}\theta_b^* + \delta_b)$ is strictly increasing in θ_b^* . $\square$

Proof of Proposition 2, Parts (i)–(ii)

Part (i). Write the LW condition as $F(q; i) \equiv i - \bar{\alpha}_h[v'(q)/g'(q) - 1] = 0$. By the implicit function theorem, $dq^*/di = -F_i/F_q$. The numerator is $F_i = 1 > 0$. For the denominator, use the Nash bargaining solution $g(q) = (1-\eta_h)v(q) + \eta_h q$ (Section 2.2), giving $g' = (1-\eta_h)v' + \eta_h$ and $g'' = (1-\eta_h)v''$. By the quotient rule:

$$F_q = -\bar{\alpha}_h \frac{d}{dq} \left[\frac{v'(q)}{g'(q)} \right] = -\bar{\alpha}_h \cdot \frac{v'' \eta_h}{[g'(q)]^2} = \frac{\bar{\alpha}_h \eta_h |v''(q)|}{[g'(q)]^2} > 0,$$

where the sign uses $v'' < 0$. Hence $dq^*/di = -[g'(q^*)]^2/(\bar{\alpha}_h \eta_h |v''(q^*)|) < 0$.

Part (ii). Let $G(u, q, \ell^*) \equiv c_v - \beta \lambda_f(u)(1-\eta_L)(\bar{R} - b)/(r + \delta_L)$. Differentiating $G = 0$ with ℓ^* fixed gives $\partial u^*/\partial q^* = -G_q/G_u$. Since $\partial \bar{R}/\partial q^* = \alpha_f(1-\eta_h)(v'(q^*) - 1) > 0$ for $q^* < q^{\text{FB}}$,

we have $G_q < 0$. $G_u < 0$ because $\lambda_f(u)$ is strictly increasing in u for $\sigma_m < 1$ (Proposition 1). Therefore $A_{\text{Liq}} \equiv (-G_q/G_u)(dq^*/di) = (\text{neg}) \cdot (\text{neg}) > 0$. $\square$

Proof of Proposition 2, Parts (iii)–(iv)

Part (iii). By Lemma 2, $\Delta_{\text{firm}}(s)$ is strictly increasing in s . The BFE (BFE) equates the fixed cost κ to $\beta q_b(\theta_b)(1 - \eta_b)\Delta_{\text{firm}}(s)$; since $q_b = \bar{\alpha}/(1 + \theta_b)$ is decreasing in θ_b , the left-hand side is constant while the right-hand side is strictly decreasing in θ_b , and strictly increasing in $\Delta_{\text{firm}}(s)$. Hence $\theta_b^*(s)$ is strictly increasing in s . By (11), ℓ^* is strictly increasing in θ_b^* , so $d\ell^*/di = \frac{1}{4} d\ell^*/ds > 0$.

Part (iv). The financial intermediation channel contribution to du^*/di is:

$$A_{\text{FI}} \equiv -\frac{G_\ell d\ell^*/di + G_s^{\text{FI}}/4}{G_u},$$

where $G_\ell \equiv \partial G/\partial \ell^* = -\beta \lambda_f(1 - \eta_L)[F(k_b) - F(k_u)]/(r + \delta_L)$ and $G_s^{\text{FI}} \equiv -\beta \lambda_f(1 - \eta_L)\ell^*[F'(k_b)dk_b/ds - F'(k_u)dk_u/ds]/(r + \delta_L)$ captures the banking-differential contribution to $\partial G/\partial s$ (the k_u component is absorbed into A_{Liq}).

For $s \leq \hat{s}$: $k_b = k_u$, so $F(k_b) = F(k_u)$ implies $G_\ell = 0$. Also $dk_b/ds = dk_u/ds$ in the low regime, so $G_s^{\text{FI}} = 0$. Hence $A_{\text{FI}} = 0$.

For $s > \hat{s}$: $k_b > k_u$ (Lemma 1) gives $G_\ell < 0$. Moreover $F'(k_b)dk_b/ds > 0 > F'(k_u)dk_u/ds$ in the high regime (since $dk_b/ds = (s/B)^{1/\gamma-1}/(\gamma B) > 0$ while $dk_u/ds < 0$), so $G_s^{\text{FI}} < 0$. Since $G_u < 0$, both numerator terms are negative, giving $A_{\text{FI}} < 0$. $\square$

B. Calibration Details

The model is calibrated quarterly. Table 1 lists all parameters; Table 2 reports the fit against the five empirical targets.

External parameters. $\beta = 0.992$ corresponds to an annualized real interest rate of 3.2%, measured as the average ex-post real return on short-term US Treasury bills over 1960–2019 (Federal Reserve H.15 release, GDP-deflator adjusted). $\delta_L = 0.050$ is the quarterly layoff-and-discharge rate from the BLS Job Openings and Labor Turnover Survey (JOLTS) averaged over 2001–2019, rounded to 5% following Berentsen et al. (2011). The matching elasticity $\sigma_m = 0.280$ is set equal to the worker bargaining weight η_L (Hosios condition); the value lies at the lower end of the range $[0.3, 0.7]$ surveyed by Petrongolo and Pissarides (2001). The matching efficiency $Z = 0.364$ is taken from Berentsen et al. (2011), calibrated to reproduce the US steady-state job-finding and vacancy-filling rates at $u^* = 5\%$. The three credit-market parameters ($\delta_b, \bar{\alpha}, \kappa$) are taken directly from Bethune et al. (2021). The quarterly relationship dissolution rate $\delta_b = 0.036$ is estimated from the Federal Reserve's

TABLE 1. Benchmark Calibration

Parameter	Value	Type	Target / Source
β	0.992	External	$r = 3.2\%/yr$
δ_L	0.050	External	JOLTS separation (2001–2019)
$\eta_L = \sigma_m$	0.280	External	Hosios (1990); PP (2001)
Z	0.364	External	Berentsen et al. (2011)
η_h	0.500	Normalization	Symmetric Nash
a_{util}	0.500	Normalization	CRRA curvature
A_{util}	0.481	Internal	$M/PY(0) = 17\%$
b	1.839	Internal	$b/\bar{R}(0) = 90\%$
c_v	0.078	Internal	$u^*(0) = 5\%$
δ_b	0.036	External	STBL; Bethune et al. (2021)
$\bar{\alpha}$	0.465	External	Bethune et al. (2021)
κ	3.000	External	Bethune et al. (2021)
η_b	0.500	Normalization	Symmetric Nash
δ_{innov}	15.16	Internal	$\ell^*(0) = 65\%$
A_F, ξ_F	1.0, 0.50	Normalization	Scale/curvature
γ	1.000	Normalization	Quadratic cost
B	0.02627	Internal	$\hat{i} = 10\%/yr$

Survey of Terms of Business Lending (STBL, conducted quarterly since 1977), which tracks the transition between relationship and arm’s-length borrowing for a rotating sample of commercial and industrial loans. $\bar{\alpha} = 0.465$ (maximum bank–firm meeting probability) and $\kappa = 3.000$ (bank entry cost) are identified in Bethune et al. (2021) from the cross-sectional share of relationship borrowers and the average time-to-match in the credit market; we adopt these values without re-estimation.

Normalizations. Bargaining weights $\eta_h = \eta_b = 0.5$ impose symmetric Nash in both DM goods and credit bargaining, following standard practice in Lagos–Wright and BRZW-type models. The CRRA coefficient $a_{\text{util}} = 0.5$ is set following Berentsen et al. (2011), generating a money-demand curvature consistent with the semi-elasticity estimates of Lucas (2000); the exact implied elasticity in the Lagos–Wright framework also depends on the bargaining weight η_h through $g'(q) = (1 - \eta_h)\nu'(q) + \eta_h$. Capital production uses $A_F = 1$ (scale) and $\xi_F = 0.5$ (curvature); only the ratio k_b^*/k_u^* matters for unemployment. The monitoring cost $\psi(d) = \frac{\beta}{2}d^2$ is quadratic ($\gamma = 1$), the simplest case generating an interior credit line.

Identification. The five internal parameters are identified by a lower-triangular block structure, all evaluated at $i = 0$. *Block I (credit market, independent of labor and goods).* $B = 0.02627$ is pinned by the regime-threshold condition $\hat{s}(1 + \hat{s})^2 = B$ at the target $\hat{i} = 10\%/yr$ ($\hat{s} = 0.025$), following Bethune et al. (2021) who associate this threshold with the peak of the Volcker disinflation (Federal Funds Rate above 10%, 1979–1982). $\delta_{\text{innov}} = 15.16$ is recovered from the BFE at $s = 0$ given the target $\ell^*(0) = 65\%$, the STBL estimate that roughly 65% of US commercial and industrial loans are extended through ongoing banking relationships (Bethune et al. 2021). *Block II (labor and goods, given Block I).* $A_{\text{util}} = 0.481$ is solved numerically from the money-demand condition to match $M/PY(0) = 17\%$, the Friedman-rule extrapolation of Lucas’s (2000) estimated US M1 demand curve (M1 divided by nominal GDP, 1900–1994). $b = 1.839$ follows from the replacement-rate target $b/\bar{R}(0) = 90\%$ after Hagedorn and Manovskii (2008), who argue that the total opportunity cost of employment—UI benefits (~40% replacement), home production, and leisure—amounts to 90–95% of the market wage. $c_v = 0.078$ is then pinned by the free-entry condition at the target $u^*(0) = 5\%$, the US civilian unemployment rate average 1960–2019 (BLS, seasonally adjusted).

All five targets are matched to high accuracy. The hump peak $i^\dagger \approx 9.9\%/yr$ and the banking fraction $\ell^*(20\%) = 86.8\%$ are untargeted predictions. The proximity of $i^\dagger$ to the calibrated threshold $\hat{i}$ reflects the channel asymmetry: the *liquidity channel* operates continuously while the *financial intermediation channel* activates only above $\hat{i}$, so the peak lies just below the regime boundary; the small gap ($i^\dagger = 9.9\%$ vs. $\hat{i} = 10\%/yr$) reflects the cash-saving margin of the *financial intermediation channel*, which is active for all $i > 0$ and

TABLE 2. Calibration Targets: Model versus Data

Moment	Model	Target	Source
$M/PY(0)$	17.0%	17%	Lucas (2000)
$b/\bar{R}(0)$	90.0%	90%	Hagedorn and Manovskii (2008)
$u^*(0)$	5.000%	5%	BLS (1960–2019)
$\hat{i}$	10.0%/yr	10%/yr	Bethune et al. (2021); Volcker peak
$\ell^*(0)$	64.7%	65%	STBL; Bethune et al. (2021)

is absorbed into A_{Liq} in the decomposition. The 22 percentage-point rise in ℓ^* from the Friedman rule to $\hat{i} = 20\%/yr$ is consistent with the upward trend in STBL relationship-lending shares documented in Bethune et al. (2021).

C. Bank Entry Congestion

The baseline model assumes a constant entry cost κ , so that bank-market tightness θ_b^* responds linearly to the match surplus $\Delta_{\text{firm}}(s)$ through the bank free-entry condition (BFE). In practice, bank entry involves regulatory compliance, credit-assessment expertise, and relationship infrastructure whose costs rise as competition for unbanked firms intensifies. This appendix relaxes the constant-cost assumption and shows that the non-monotone unemployment locus is robust.

The modified bank free-entry condition replaces κ with $\kappa(\theta_b) \equiv \kappa_0 + \kappa_1 \theta_b^\nu$:

$$(C1) \quad \kappa(\theta_b) = \beta q_b(\theta_b) (1 - \eta_b) \Delta_{\text{firm}}(s),$$

where $\kappa_0 > 0$ is the baseline cost, $\kappa_1 \geq 0$ the congestion intensity, and $\nu > 0$ the curvature. When $\kappa_1 = 0$, (C1) reduces to (BFE). When $\kappa_1 > 0$, the left-hand side is strictly increasing in θ_b and the right-hand side is strictly decreasing, so a unique solution $\theta_b^*(s) \geq 0$ exists whenever $\beta \bar{\alpha}(1 - \eta_b) \Delta_{\text{firm}}(s) > \kappa_0$. Congestion dampens bank entry. For any given $\Delta_{\text{firm}}(s)$, the term $\kappa_1 \theta_b^\nu$ raises the marginal cost of entry, lowering θ_b^* and hence ℓ^* relative to the constant-cost benchmark. Although the absolute reduction in θ_b^* is largest at high spreads (where θ_b^* is large), the proportional effect on ℓ^* is largest at the Friedman rule: there ℓ^* is not yet near the matching saturation ceiling, so entry reductions translate more fully into lower banking fractions.

Figure 7 plots $\ell^*(i)$ and $u^*(i)$ for $\kappa_1 \in \{0, 0.5, 1.5, 5\}$ with $\nu = 1$. The left panel shows that congestion substantially reduces the banking fraction: with $\kappa_1 = 5$, $\ell^*(0)$ falls from 64.7% to 42.2%, and $\ell^*(20\%)$ from 86.7% to 75.8%. The right panel reveals a striking robustness result. Despite these large differences in banking fractions, the unemployment locus is nearly invariant to κ_1 . The hump peak remains at $u^* = 5.54\%$ near $\hat{i} = 10\%/yr$ for all values of κ_1 ; the post-peak decline shifts by only 0.13 percentage points at $i = 20\%/yr$.

The insensitivity of unemployment to congestion has a structural explanation. The banking fraction enters the match surplus $\bar{R}$ only through the capital-output component $\ell^*[F(k_b) - F(k_u)]$; the goods-market component $\alpha_f(1 - \eta_h)[v(q) - q]$ is independent of ℓ^* . Congestion affects the *level* of ℓ^* but not the threshold $\hat{s}$: the hump shape is driven by the discrete activation of the capital-investment margin at $\hat{s}$, which is κ_1 -independent. The non-monotone locus is therefore robust to congestion in the credit market.

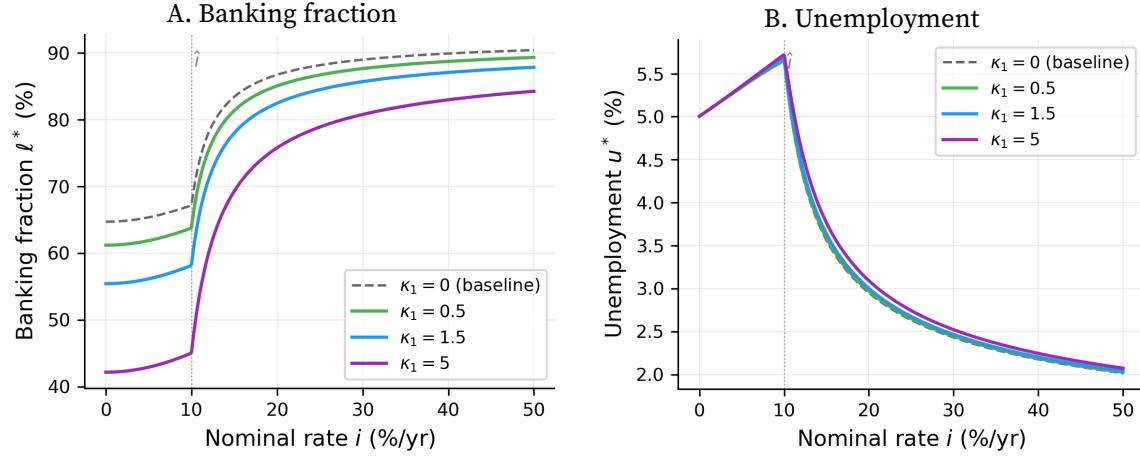


FIGURE 7. Bank entry congestion.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the author(s) used Claude (Anthropic) in order to assist with editing and refining the manuscript text, improving language and readability, and supporting literature review and summarization. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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